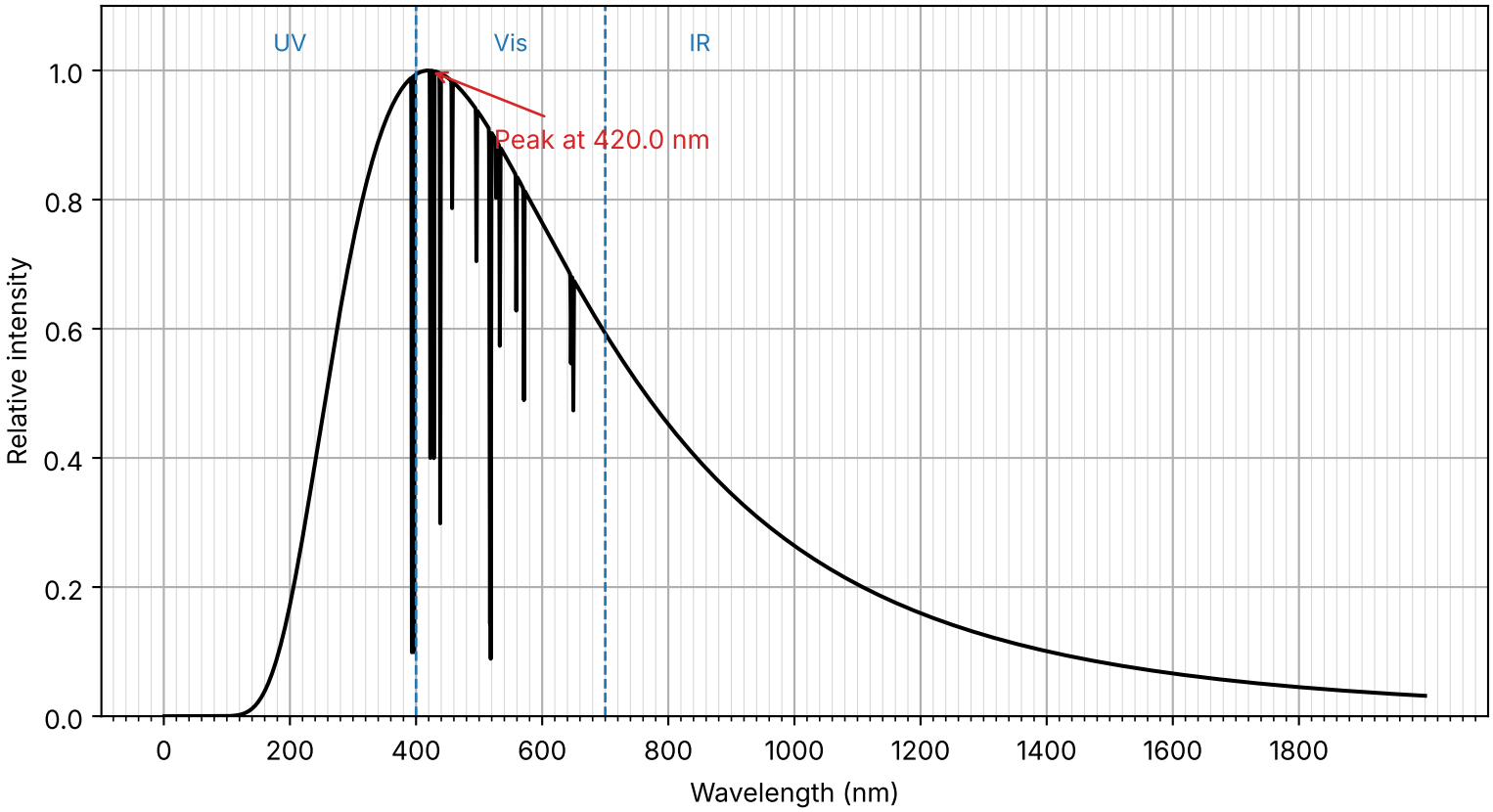
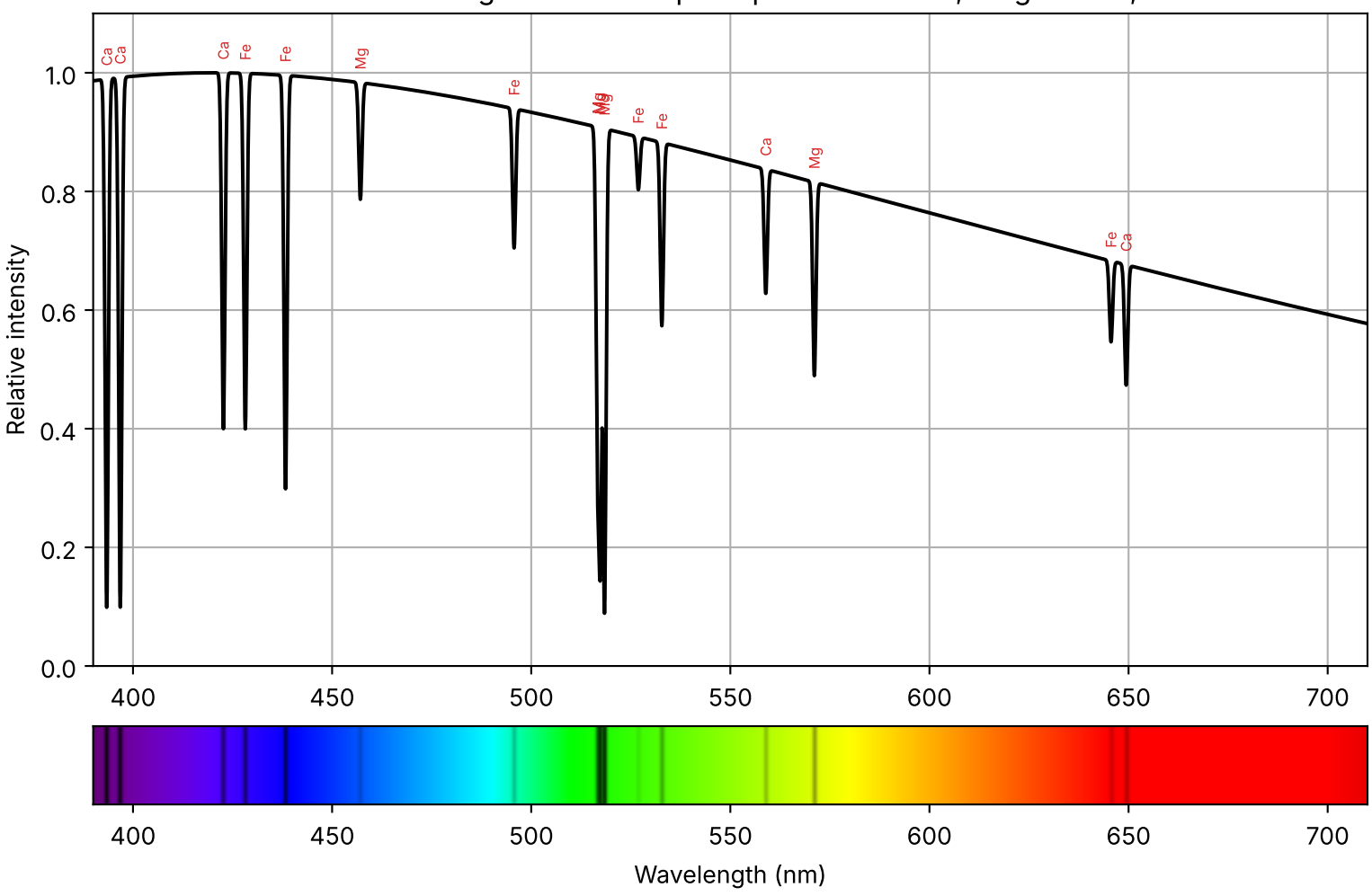
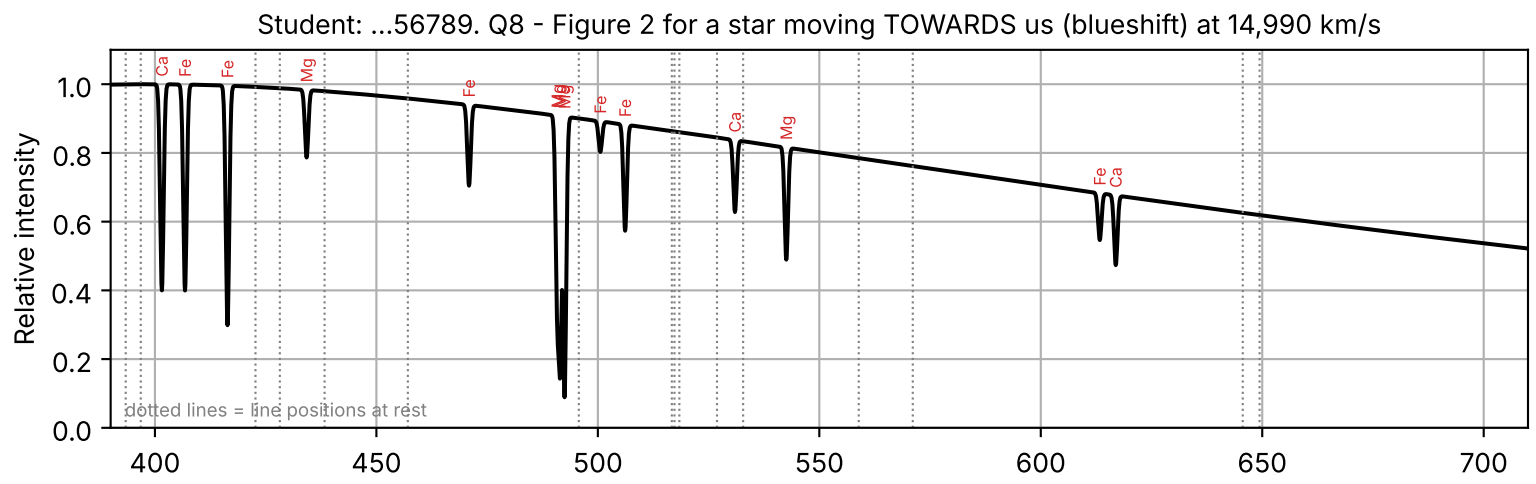
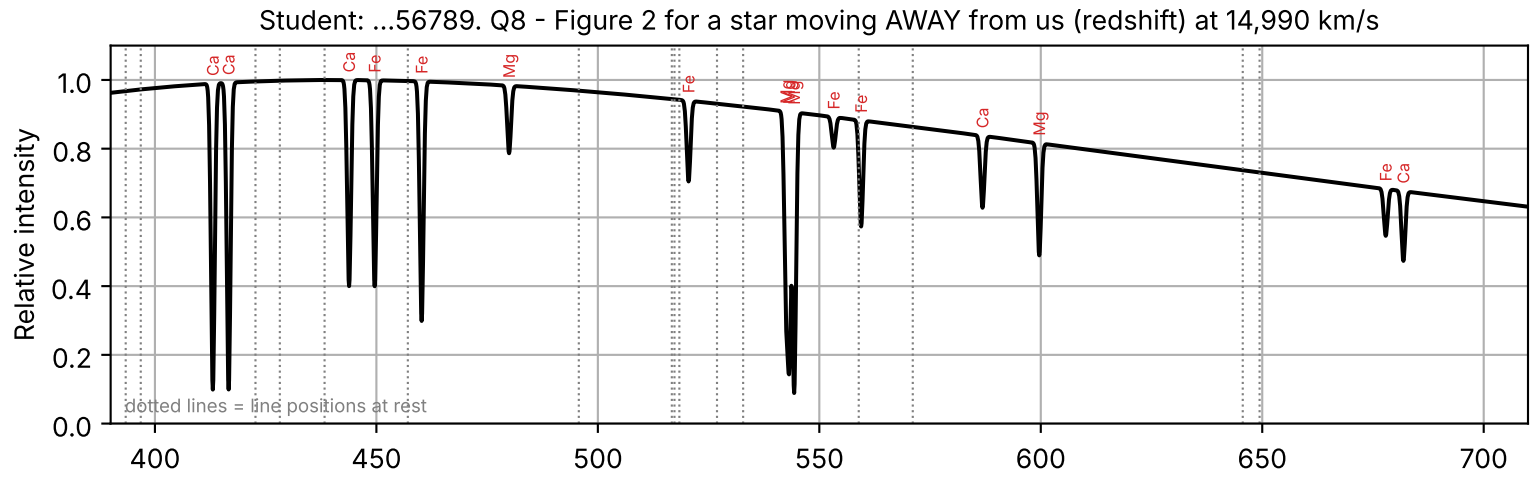
Student: ...56789. Figure 1 - Spectrum at 6900 K (type F)

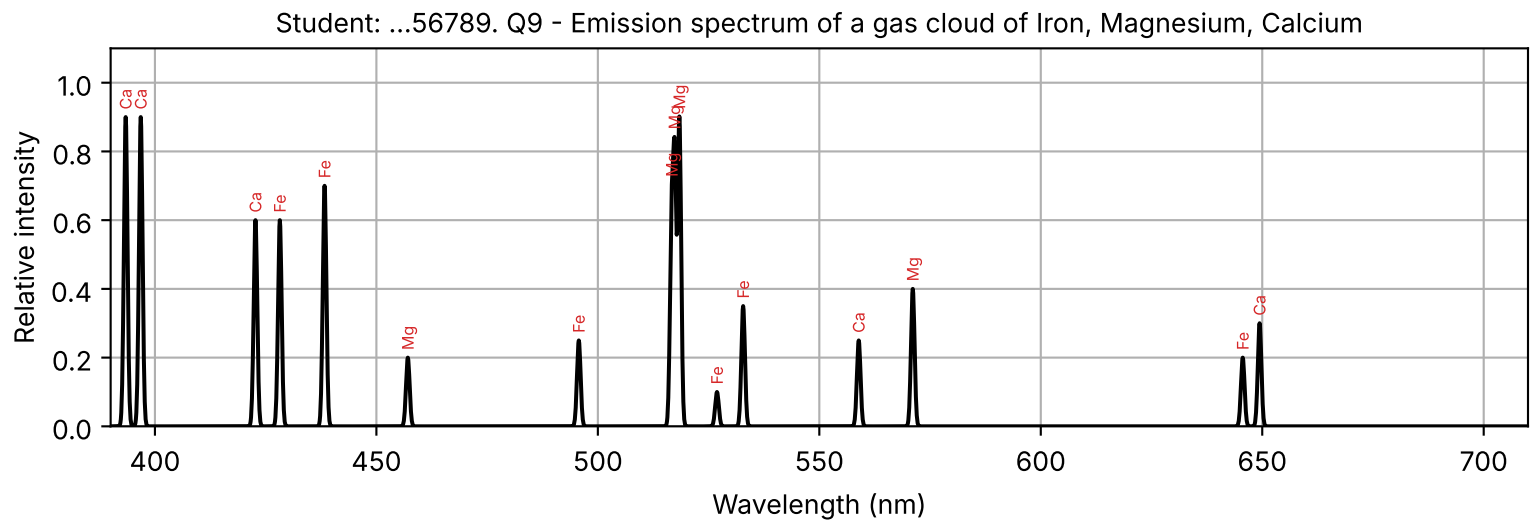


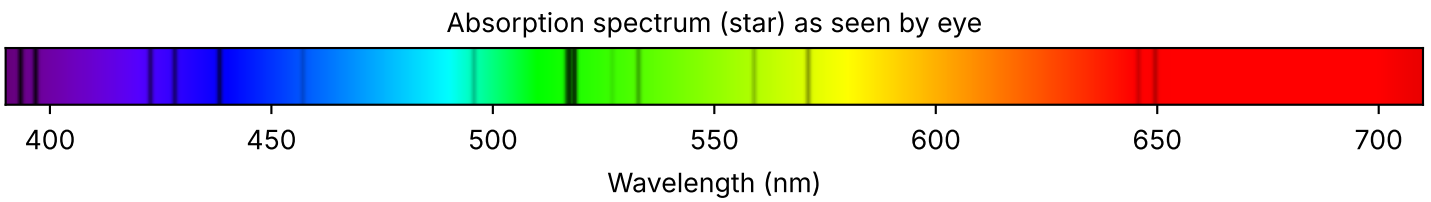
Student: ...56789. Figure 2 - Absorption pattern for Iron, Magnesium, Calcium

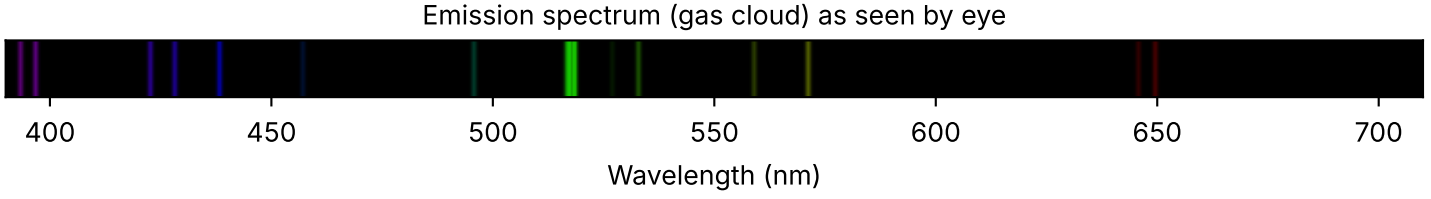



Student: ...56789. Q8 - Figure 2 for a star moving TOWARDS us (blueshift) at 14,990 km/s
Relative intensity
dotted lines = line positions at rest
Ca
Fe
Fe
Mg
Fe
Mg
Mg
Fe
Fe
Ca
Mg
Fe
Ca
0.0
0.2
0.4
0.6
0.8
1.0
400
450
500
550
600
650
700

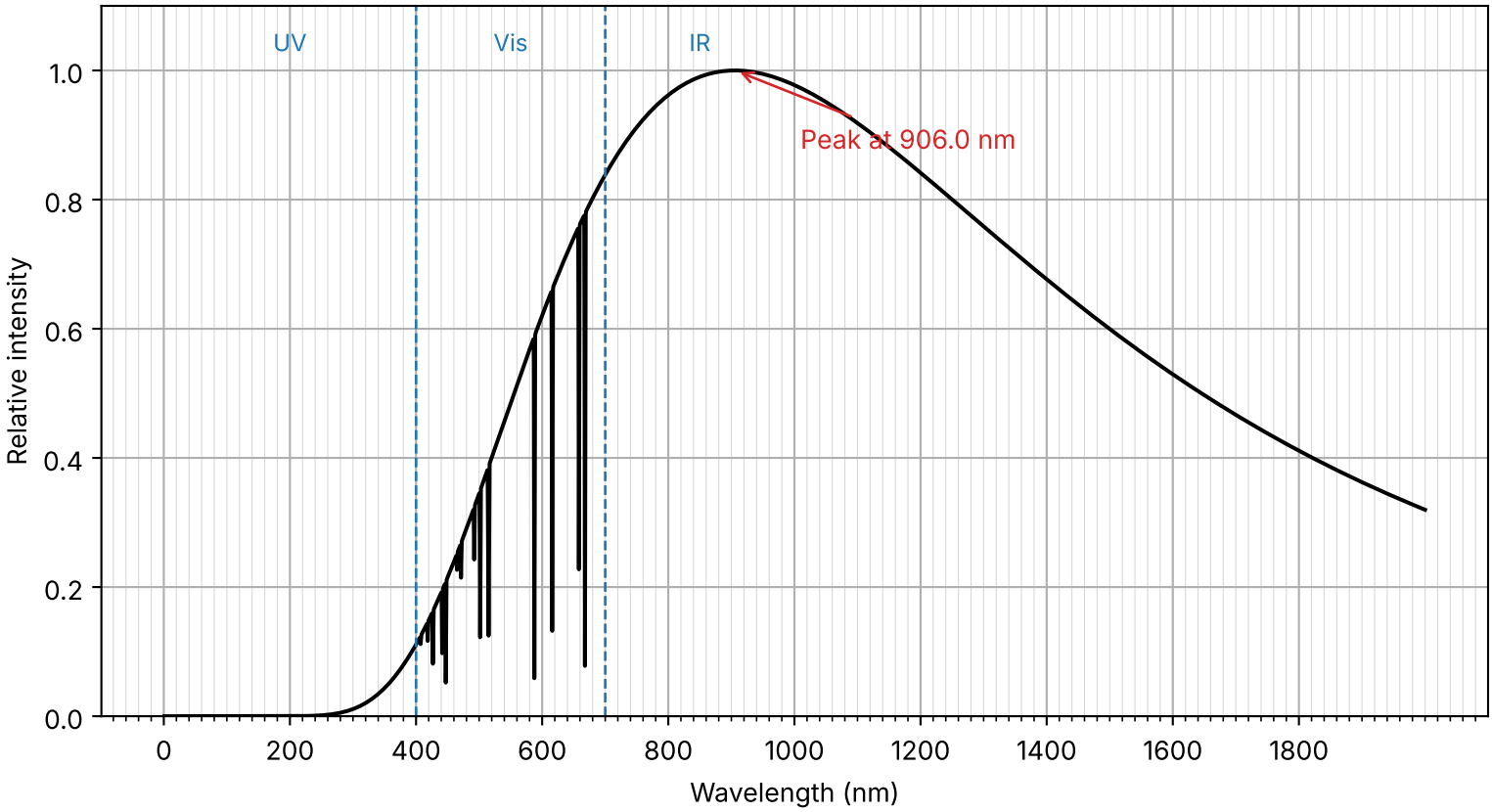

Student: ...56789. Q8 - Figure 2 for a star moving AWAY from us (redshift) at 14,990 km/s
Relative intensity
dotted lines = line positions at rest
Ca
Ca
Ca
Fe
Fe
Mg
Fe
Mg
Mg
Fe
Fe
Ca
Mg
Fe
Ca
0.0
0.2
0.4
0.6
0.8
1.0
400
450
500
550
600
650
700


Student: ...56789. Q9 - Emission spectrum of a gas cloud of Iron, Magnesium, Calcium
Relative intensity
Ca
Ca
Ca
Fe
Mg
Fe
Mg
Mg
Mg
Fe
Fe
Ca
Mg
Fe
Ca
0.0
0.2
0.4
0.6
0.8
1.0
400
450
500
550
600
650
700
Wavelength (nm)

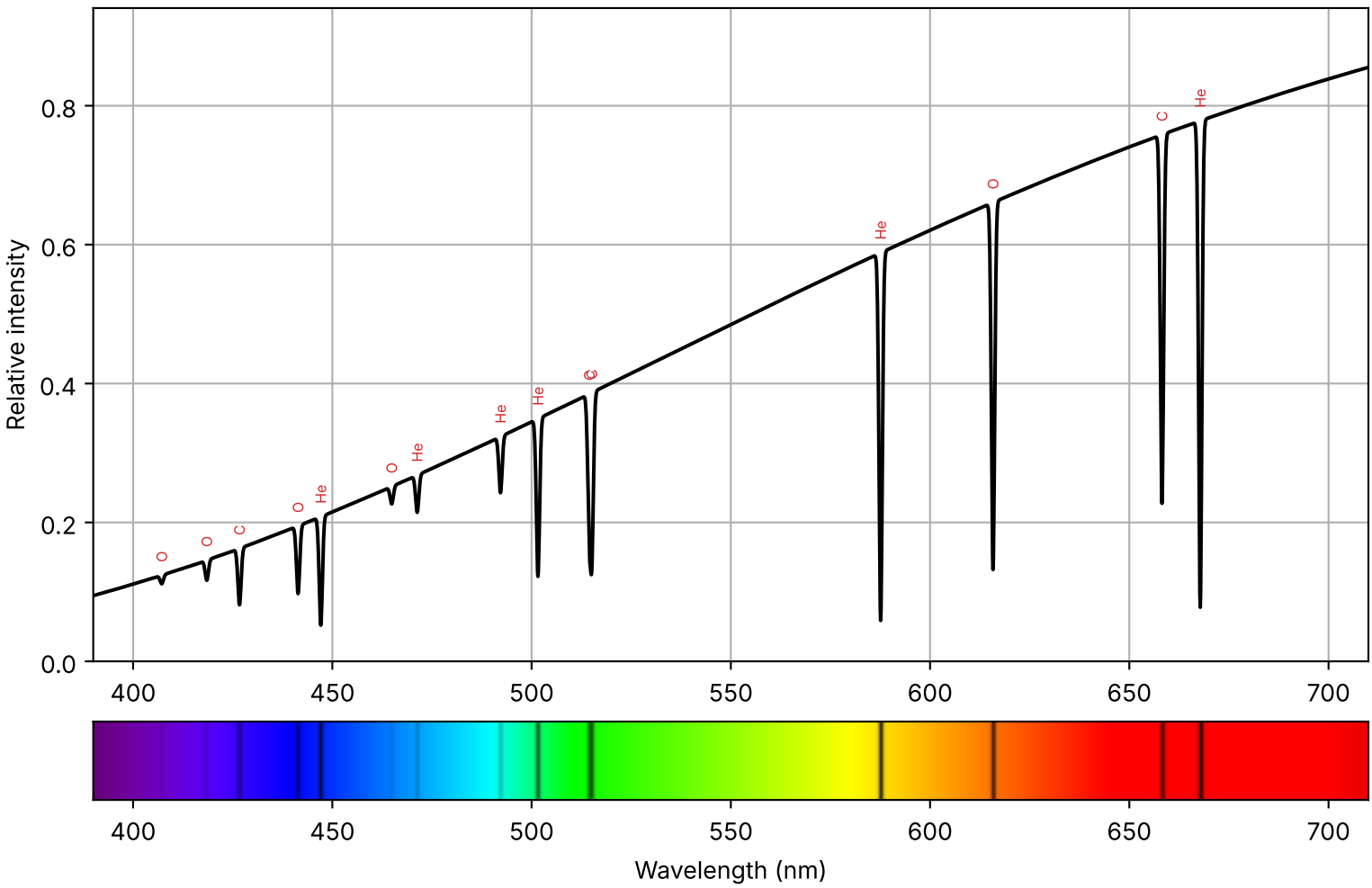

Absorption spectrum (star) as seen by eye
400
450
500
550
600
650
700
Wavelength (nm)

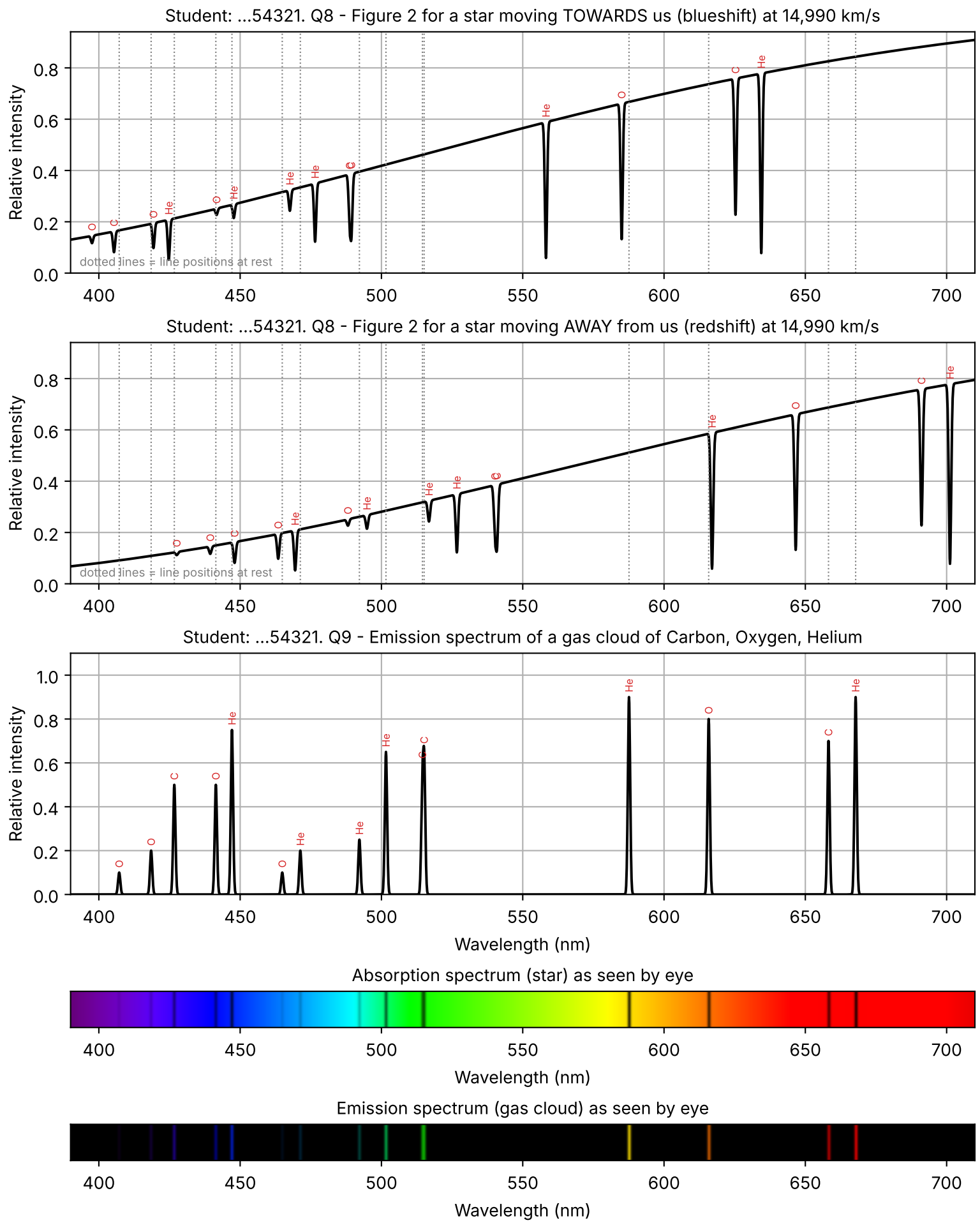

Emission spectrum (gas cloud) as seen by eye
400
450
500
550
600
650
700
Wavelength (nm)

Student: ...54321. Figure 1 - Spectrum at 3200 K (type M)



Student: ...54321. Figure 2 - Absorption pattern for Carbon, Oxygen, Helium




Student: ...54321. Q8 - Figure 2 for a star moving TOWARDS us (blueshift) at 14,990 km/s
Relative intensity
dotted lines = line positions at rest
Student: ...54321. Q8 - Figure 2 for a star moving AWAY from us (redshift) at 14,990 km/s
Relative intensity
dotted lines = line positions at rest
Student: ...54321. Q9 - Emission spectrum of a gas cloud of Carbon, Oxygen, Helium
Relative intensity
Wavelength (nm)
Absorption spectrum (star) as seen by eye
Wavelength (nm)
Emission spectrum (gas cloud) as seen by eye
Wavelength (nm)